Contact Information

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Education

2012–2017	**Ph.D. in Mathematics**, New York University Advisors: Erwin Lutwak, Deane Yang, and Gaoyong Zhang Thesis: Geometric measures, affine invariants, and their characterizations
2007–2011	**B.S. in Mathematics**, Shanghai University

Research Interests

convex geometry and related partial differential equations

Employment

2021 –	Assistant Professor at Syracuse University
2018 – 2021	C.L.E. Moore Instructor at Massachusetts Institute of Technology Mentor: David Jerison
2017 – 2018	Assistant Professor (Contract Faculty) at St. John's University
Summer 2017	Research Associates & Adjunct Assistant Professor at New York University

Grant

- NSF CAREER Grant DMS-2337630 (sole PI), 06/2024 – 05/2029, $434,697.00
- NSF Grant DMS-2002778 (sole PI, transferred from DMS-2132330), 06/2020 – 05/2024, $159,159.00

Publications and Preprints

It is important to note that the convention in pure math, as documented by the American Mathematical Society, is to list authors in alphabetical order.

1. (with D. Xi) Geometric measures for radial Minkowski homomorphisms and Brunn-Minkowski-type inequalities, submitted to *Mathematische Annalen*, April 2026.
2. (with D. Xi) Fractional affine area measures, submitted to *Journal of the European Mathematical Society* in Dec 2024, revised and resubmitted in Dec 2025.
3. (with V. Semenov) The growth rate of surface area measure for noncompact convex sets with prescribed asymptotic cone, *Transactions of the American Mathematical Society*, 378 (11): 7945-7975, 2025.
4. (with D. Langharst and M. Roysdon) On the m-th order affine Pólya-Szegö principle, *Journal of Geometric Analysis*, 35 (7): Paper No. 205, 32pp, 2025.
5. (with N. Fang, D. Ye, and Z. Zhang) Dual Orlicz curvature measures for log-concave functions and their Minkowski problems, *Calculus of Variations and Partial Differential Equations*, 64: Paper No. 44, 31pp, 2025.
6. (with Y. Huang, J. Liu, and D. Xi) Dual curvature measures for log-concave functions, *Journal of Differential Geometry*, 128 (2): 815-860, 2024.
7. (with L. Guo and D. Xi) The L_p chord Minkowski problem in a critical interval. *Mathematische Annalen*, 389: 3123-3162, 2024.
8. (with S. Chen, S. Hu, and W. Liu) On the planar Gaussian-Minkowski problem, *Advances in Mathematics*, Volume 435, Part A: 109351, 32pp, 2023.
9. (with D. Xi, D. Yang, and G. Zhang) The L_p chord Minkowski problem, *Advanced Nonlinear Studies*, 23: pp. 20220041, 2023.
10. (with D. Xi) General affine invariances related to Mahler volume, *International Mathematics Research Notices*, 2022: 14151-14180, 2022.
11. (with Y. Huang and D. Xi) The Minkowski problem in Gaussian probability space, *Advances in Mathematics*, 385: Paper No. 107769, 36 pp, 2021.

12. (with K. Böröczky, E. Lutwak, D. Yang, and G. Zhang) The Gauss image problem, *Communications on Pure and Applied Mathematics*, 73: 1406-1452, 2020.

13. (with K. Böröczky, E. Lutwak, D. Yang, and G. Zhang) The dual Minkowski problem for symmetric convex bodies, *Advances in Mathematics*, 356:106805, 2019.

14. The L_p Aleksandrov problem for origin-symmetric polytopes, *Proceedings of the American Mathematical Society*, 147 (10): 4477-4492, 2019.

15. (with C. Chen, and Y. Huang) Smooth solutions to the L_p-dual Minkowski problem, *Mathematische Annalen*, 373 (3-4):953-976, 2019.

16. (with Y. Huang) On the L_p dual Minkowski problem, *Advances in Mathematics*, 332: 57-84, 2018.

17. Existence of solutions to the even dual Minkowski problem. *Journal of Differential Geometry*, 110 (3): 543-572, 2018.

18. Geometric measures, affine invariants, and their characterizations. *Ph.D. thesis*, New York University, 2017.

19. The dual Minkowski problem for negative indices. *Calculus of Variations and Partial Differential Equations*, 56 (2):18, 2017.

20. On L_p-affine surface area and curvature measures. *International Mathematics Research Notices*, (5): 1387–1423, 2016.

Invited Talks

- 2026 Apr. The online Geometric and Functional Inequalities and Applications seminar: SL(n)-invariant isoperimetric and Minkowski problems.

- 2026 Mar. Binghamton University, Analysis Seminar: SL(n)-invariant isoperimetric and Minkowski problems.

- 2025 Oct. AMS Fall Central Sectional Meeting at St. Louis: Affine area measures.

- 2024 Dec. MFO–Oberwolfach workshop “Convex Geometry and its Applications” at Oberwolfach, Germany: Growth rate of surface area measures for C-asymptotic sets.
- 2024 May. Convex Geometry and Related PDEs conference at Changsha, China: On the geometry of log-concave functions.
- 2023 Dec. Winter meeting of Canadian Mathematical Society at Montreal: The Minkowski problem in Gaussian probability space.
- 2023 Dec. Beijing Technology and Business University (virtual): The Minkowski problem in Gaussian probability space.
- 2023 Jun. INdAM Meeting “Convex Geometry - Analytic Aspects” at Cortona, Italy: The Minkowski problem in Gaussian probability space. (**main speaker**)
- 2023 Jun. Summer Meeting of Canadian Mathematical Society at Ottawa: The Minkowski problem in Gaussian probability space.
- 2023 May. Southwest University (virtual): The dual Minkowski problem. (3-hour short lecture)
- 2023 May. Chinese Academy of Sciences (virtual): The Minkowski problem in Gaussian probability space.
- 2022 Dec. Shanghai University (virtual): Dual curvature measures for log-concave functions.
- 2022 Dec. Beihang University (virtual): Dual curvature measures for log-concave functions.
- 2022 Dec. Shanxi Normal University (virtual): Dual curvature measures for log-concave functions.
- 2022 Nov. Cornell University: Dual curvature measures for log-concave functions.
- 2022 Sep. ICERM: Dual curvature measures for log-concave functions.
- 2021 Sep. Syracuse University, Analysis Seminar: Mass transport problem on the unit sphere via Gauss map.

- 2021 May Jilin Normal University (virtual): Mass transport problem on the unit sphere via Gauss map.
- 2021 Mar. Syracuse University (virtual, colloquium): Recovering the shapes of convex bodies.
- 2021 Feb. Florida International University (virtual, colloquium): Recovering the shapes of convex bodies.
- 2021 Feb. University of Georgia (virtual, colloquium): Recovering the shapes of convex bodies.
- 2021 Jan. UC San Diego (virtual): Mass transport problem on the unit sphere via Gauss map.
- 2021 Jan. UC Santa Cruz (virtual, colloquium): Recovering the shapes of convex bodies.
- 2020 Oct. AMS special session (virtual): The Minkowski problem in Gaussian probability space.
- 2020 Aug. University of Connecticut (virtual): Reconstruction of convex bodies via Gauss map.
- 2019 Jun. International Congress of Chinese Mathematicians, 45-min talk: The dual Minkowski problem for o-symmetric convex bodies.
- 2019 Jun. Fudan University: The dual Minkowski problem for o-symmetric convex bodies.
- 2019 Jun. Tongji University: The dual Minkowski problem for o-symmetric convex bodies.
- 2019 Jun. Shanghai University: The dual Minkowski problem for o-symmetric convex bodies.
- 2019 Jun. Hunan University, lecture series: An Introduction to Minkowski-type problems in convex geometry.
- 2019 May. AIM workshop: The even dual Minkowski problem for integer indices.

- 2019 Jan. University of Connecticut, PDE and Differential Geometry Seminar: The Gauss image problem.
- 2018 Mar. AMS special session at Ohio State University: The Aleksandrov problem and its recent development.
- 2017 Dec. St. John's University: Minkowski problems and Monge-Ampère type equations.
- 2017 Sept. CUNY Graduate Center, Geometric Analysis Seminar: Minkowski-type problems in convex geometry.
- 2017 Feb. Case Western Reserve University, Analysis & Probability Seminar: On the dual Minkowski problem.
- 2017 Feb. Kent State University, Measure Theory Seminar: The dual Minkowski problem and its solution.
- 2015 Sep. Oaxaca, Mexico (CMO workshop): On L_p-affine surface area and curvature measures.

Courses Taught

- Syracuse University

Spring	2026:	MAT-602 Fundamentals of Analysis II
Fall	2025:	MAT-296 Calculus II
		MAT-512 Introduction to Real Analysis II
Spring	2025:	MAT-412 Introduction to Real Analysis I
		MAT-800 Topics in Analysis
		An introduction to convex geometric analysis
Fall	2024:	on research leave
Spring	2024:	MAT-602 Fundamentals of Analysis II
Fall	2023:	MAT-412 Introduction to Real Analysis I
Spring	2023:	MAT-602 Fundamentals of Analysis II
Fall	2022:	MAT-512 Introduction to Real Analysis II
Spring	2022:	MAT-296 Calculus II
Fall	2021:	MAT-296 Calculus II

- Massachusetts Institute of Technology

Spring 2021:	18.02 Calculus (recitation leader)
Fall 2020:	18.100Q Communication Intensive Real Analysis
Spring 2020:	18.03 Differential Equations (recitation leader)
Fall 2019:	18.100Q Communication Intensive Real Analysis
Spring 2019:	18.03 Differential Equations (recitation leader)
Fall 2018:	18.01A/18.02A Calculus (recitation leader)

- St. John's University

Spring 2018:	MTH-1260 Pharmacy Calculus; MTH-1320 Business Calculus
Fall 2017:	MTH-1250 Pharmacy Statistics; MTH-1300 College Algebra

Professional Services

- Grant review

- NSF panelist, 2025–2026
- NSF ad-hoc reviewer, 2025

- Committee services

- Graduate Committee, Syracuse University, Fall 2025 – present
- External Ph.D. thesis examiner for Xia Zhou, Memorial University of Newfoundland, Canada, Summer 2025
- External Ph.D. thesis examiner for Wen Ai, Memorial University of Newfoundland, Canada, Spring 2025
- Ph.D. thesis defense committee member for Jesse Hulse, Syracuse University, Spring 2025
- Applied Math hiring committee, Syracuse University, Fall 2023
- Probability hiring committee, Syracuse University, Fall 2022

- Conference/workshop organization

- Co-organizer of the *Algebra Workshop for MAT-295*, Syracuse University, Fall 2025 – present.

  In recent years, there has been a nation-wide trend of more students struggling with Calculus courses due to weak background in algebra. Literature shows that a remedial prerequisite course in algebra might not be particularly effective.

  This workshop is designed to provide students enrolled in MAT-295 Calculus I at Syracuse University with *just-in-time* algebra support. The idea is to have students brush up their relevant algebra skills to be used in the next Calculus Lecture. The workshop is co-organized by Pawel Grzegrzolka and myself, led by an undergraduate student, and funded by my NSF CAREER grant.

- Co-organizer of the *Analysis Seminar*, Syracuse University, March 2025 – present.

- Co-organizer of the AMS Special Session on *Probabilistic and Analytic Aspects in Convexity*, October 2024.

- Co-organizer of the Special Session *Interplay Between Analysis and Convexity* in the 2023 Summer Meeting of the Canadian Mathematical Society

- **Journal referee**

  I refereed/provided quick opinion *multiple times* for the following journals:

  – Advances in Applied Mathematics
  – Advances in Geometry
  – Advances in Mathematics
  – Advanced Nonlinear Studies
  – Bulletin des sciences mathématiques
  – Calculus of Variations and Partial Differential Equations
  – Discrete and Continuous Dynamical Systems Series S
  – Indagationes Mathematicae
  – International Mathematics Research Notices

  - Journal of Differential Geometry
  - Journal of Differential Equations
  - Journal of Geometric Analysis
  - Journal of Mathematical Analysis and Applications
  - Journal of the European Mathematical Society
  - Mathematische Annalen
  - Nonlinear Analysis
  - Proceedings of the American Mathematical Society
  - Rocky Mountain Journal of Mathematics
  - SCIENCE CHINA Mathematics
  - Transactions of the American Mathematical Society

  I also write reviews for *MathSciNet* and *zbMATH*.

- Service to the community

- Professional development mini-course for local high school teachers, *From Shape Discovery to Daily Life: bees cats, engineering, medicine, ...*, October–November, 2025.

  I developed and taught a mini-course for local high school teachers in the central NYS area.

  This 6-hour mini-course was attended by around 25 local high school teachers. Most of them were recruited from the CNY Master Teacher Program, except for two recruited by myself. This mini-course conveys my research to high school teachers in an accessible way. The goal is to show the teachers (and consequently their students) that Mathematics is much more than a set of formulas, and that just like any other sciences, one can explore, discover, and experiment in Mathematics.

  This mini-course was well-received by the teachers based on a survey conducted after the course.

- Presenter of the "Be the Scientist" program at the *Milton J. Rubenstein Museum of Science and Technology* (MOST) in Syracuse, NY, May, 2025.

- Together with Dr. Ruth Chen and Cory Monahan, we designed and supervised a community outreach program that provided Grade 2–8 students at *the North Side Learning Center* weekly fun problems in Mathematics. The goal of this program is to expose the fun and exploratory side of Mathematics to students early in their educational career. This program took place during the months of July and August, 2024.

Graduate student advising

- Cory Monahan (Ph.D. student), since 2024;
- Xinfa Meng (Ph.D. student), informally since 2024.

Post-doc mentoring

- Auttawich Manui (will arrive at Syracuse University in Fall 2026);
- Sudan Xing, 2023, currently Assistant Professor at University of Arkansas at Little Rock.